

Adham Ahmed

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Experience

halan (First private unicorn in Egypt) Jun. 2024 – Present

Software Engineer

Go, Scala, MongoDB, Kafka, Kubernetes, Redis, Elasticsearch

- Led the backend for the **Halan Investments** platform, managing systems that hold over **3 Trillion EGP** in equity funds and resolving critical production issues.
- Optimized a core service to handle 50 concurrent requests per pod instead of just one, **increasing throughput by 50x** and significantly reducing server costs.
- Built a microservices flow to handle trading orders, integrating securely with third-party APIs like the **Egyptian Bourse**.
- Designed and engineered a **Server-Driven UI (SDUI)** architecture, allowing the backend to dynamically update iOS and Android app layouts without requiring new mobile app updates.
- Developed the **Secured Loan Lock** feature, allowing users to lock their equity funds and use their investments as proof of income to get credit balances.
- Managed new features from initial product requirements to production, handling system design and delegating tasks to the engineering team.
- Refactored the investment cash-in service to integrate with Halan's **Unified Payment Gateway (UPG)**, overseeing staging, testing, and deployment.
- Delivered key features for the successful production launch of the **Halan Wallet**, fulfilling all technical requirements for **ABK Bank**.
- Engineered the **Microfinance Profile** module, leading to a **40% increase** in users viewing their installment plans and an **88% increase** in in-app installment payments.
- Helped migrate the loan and installment payment service from Scala to Go, moving to an event-driven **Kafka** architecture to improve communication with the core banking system.

bld.ai Nov. 2022 - Nov. 2023

Software Engineer

Python, Django, Git, GitLab, Open-AI, React.js, Docker

- Worked on an internal landing page project at bld.ai, utilizing ChatGPT API to generate dynamic content from Google Sheets data.
- Collaborated across teams to align project objectives, enhancing user engagement.

bld.ai Jul. 2022 - Oct. 2022

Software Engineering Intern

Python, JavaScript, HTML, CSS, Bootstrap, Django, React.js

- Completed a full-stack internship at bld.ai, gaining proficiency in frontend development with React.js and backend development using Django and REST APIs.

Orange Innovation Nov. 2021 - May. 2022

Software Engineering Intern

Java, SQL, Spring, Spring Boot, Threads

- Contributed to the development of a production project involving a chatbot deployed on Smartly.ai, built using Spring Boot for the backend. The chatbot engaged users by suggesting healthy food recipes and provided YouTube videos for the chosen recipe YummyBot.
- Learned Java sockets, JDBC, and threads.

Education

October 6 University (O6U) Sep. 2019 - Jul. 2023

B.S. in Computer Science

Achieved an Excellent Degree for my graduation project.

Projects

E-learning Management System (ELMS) Feb. 2023 - Ongoing

E-learning Website Meeting, Assignments & Quizzes

Python, JavaScript, Django, DRF & channels, React.js

- Developed a student-to-student connectivity feature using Django Channels.
- Integrated Google Calendar API to automate event scheduling and notifications for platform users.
- Implemented features for assignment submission and quizzes within the platform.

Biometric Attendance	Oct. 2023 - Ongoing
Desktop Application project to create reports and Calculate salaries.	<i>Python, Tkinter, OpenPyXL</i>
• Desktop application with a user interface to collect employee data and attendance events from the ZK-14 Biometric Machine. • Generate attendance and payroll reports for employees in a company.	

Honors, Awards and Competitions

• Participated at Arab African Collegiate Programming Contest (2022) • Participated at Egyptian Collegiate Programming Contest (2021, 2022, 2023) • Solved +2000 problems on various online judges	
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Leadership and Volunteering

ICPC O6U Community	Sep. 2020 - present
Co-founder & Mentor	
• Guiding a community of CS students in their problem solving and Competitive Programming skills. • Orchestrating coding sessions, and coding competition events to enhance skills and encourage continuous learning.	

Skills

Languages:	
Go, Scala, Python, Java, C/C++, SQL, HTML, JavaScript	
Technologies & Tools:	
Django, DRF, Channels, React.js, Spring, Spring Boot, Git, GitLab, Linux, Docker, Kubernetes, Kafka, MongoDB, Redis, Elasticsearch, CI/CD	
Concepts:	
OOP, Problem Solving, Algorithms, Data Structures, Testing, Microservices, Event-Driven Architecture, Concurrency, REST APIs.	